

Topic to discuss

Euler-Maclaurin's Integration
Numerical Problem

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Euler-Maclaurin Integration:

The Euler-Maclaurin formula is used to improve the value obtained from the trapezoidal rule by adding a correction term involving derivatives.

It connects,

→ integration

→ summation

→ derivatives at the end points.

Formula :

$$I = \int_a^b f(x) dx$$

and the interval $[a, b]$ is divided into n equal parts
then

$$h = \frac{b-a}{n}$$

$$\begin{aligned} \int_a^b f(x) dx = & \frac{h}{2} [y_0 + 2(y_1 + y_2 + y_3 + \dots + y_{n-1}) + y_n] - \\ & \frac{h^2}{12} [y'_n - y'_0] + \frac{h^4}{720} [y_n''' - y_0'''] - \\ & \frac{h^6}{30240} [y_n^{(6)} - y_0^{(6)}] \end{aligned}$$

Numerical Problem

Q: Evaluate $\int_0^{\pi/2} \sin x dx$ using Euler-Maclaurin Integration.

Solution: To solve it simply, taking $n=2$

$$\therefore h = \frac{\pi/2 - 0}{2} = \frac{\pi}{4}$$

Derivatives of $f(x)$,

$$y = f(x) = \sin x$$

$$y' = f'(x) = \cos x$$

$$y'' = f''(x) = -\sin x$$

$$y''' = f'''(x) = -\cos x$$

Now,

$$y' = f'(x) = \cos x$$

$$y'_n = f'(x_n) = \cos \frac{\pi}{2} = 0$$

$$y'_0 = \cos 0 = 1$$

2

$$y''' = f'''(x) = -\cos x$$

$$y'''_n = -\cos \frac{\pi}{2} = 0$$

$$y'''_0 = -\cos 0 = -1$$

To apply trapezoidal rule, let make table for $f(x) = \sin x$

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$
$f(x)$	0	0.7071	1

Now applying Euler-Maclaurin formula,

$$\int_a^b f(x) dx = \frac{h}{2} [y_0 + 2(y_1 + y_2 + y_3 + \dots + y_{n-1}) + y_n] - \frac{h^2}{12} [y'_n - y'_0] + \frac{h^4}{720} [y_n^{(4)} - y_0^{(4)}]$$

$$= \frac{\pi}{4 \times 2} [0 + 1 + 2 \times 0.7071] - \frac{\pi^2}{4^2 \times 12} [0 - 1] + \frac{\pi^4}{4^4 \times 720} [0 - (-1)]$$

$$= 0.948054 + 0.051404 + 0.0005284$$

$$= 0.999986 \text{ Ans.}$$

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